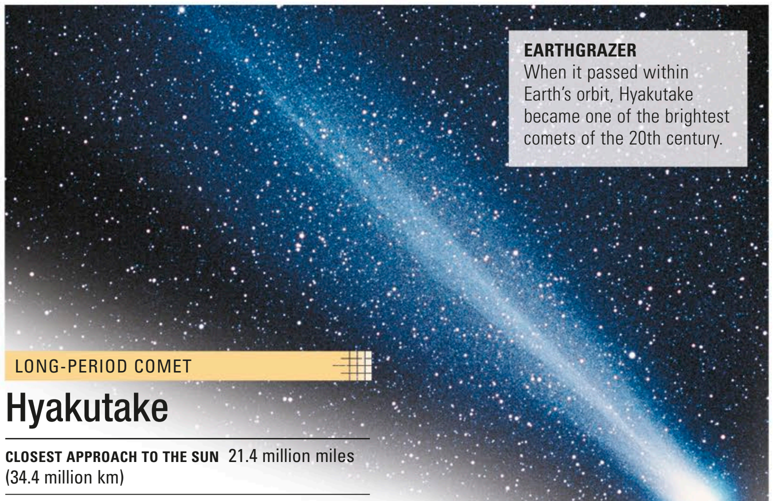


EARTHGRAZER
When it passed within Earth's orbit, Hyakutake became one of the brightest comets of the 20th century.

LONG-PERIOD COMET

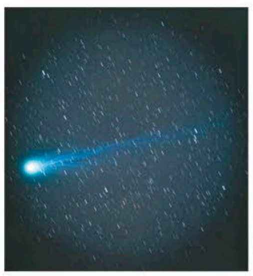
Hyakutake

CLOSEST APPROACH TO THE SUN 21.4 million miles (34.4 million km)

ORBITAL PERIOD About 30,000 years

FIRST RECORDED January 30, 1996

This comet became a Great Comet, not (like Hale–Bopp) because the nucleus was big, but because on March 24, 1996, it came within a mere 9 million miles (15 million km) of Earth. It was discovered by the Japanese amateur astronomer Hyakutake Yuji using only a pair of high-powered binoculars. The comet became so bright that large radio-telescope spectrometers could detect minor constituents in the coma, such as a compound of water and deuterium (HDO) and methanol (CH_3OH). Hyakutake was the first comet to be observed to emit X-rays. Subsequently, it was found that other comets are also sources of X-rays, the rays being produced when electrons in the coma are captured by ions in the solar wind. On May 1, 1996, the Ulysses spacecraft detected Hyakutake's gas tail when 355 million miles (570 million km) from the nucleus. This is the longest comet tail to be detected to date. Sections of Hyakutake's gas tail have disconnected due to interactions between magnetic fields in the solar wind and the tail.



TELESCOPIC VIEW
In March and April 1996, superb short-exposure photographs of Hyakutake could be obtained using only large telephoto lenses or small telescopes.

SHORT-PERIOD COMET

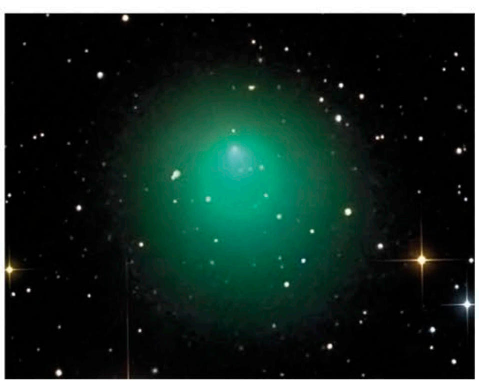
Encke

CLOSEST APPROACH TO THE SUN 32 million miles (51 million km)

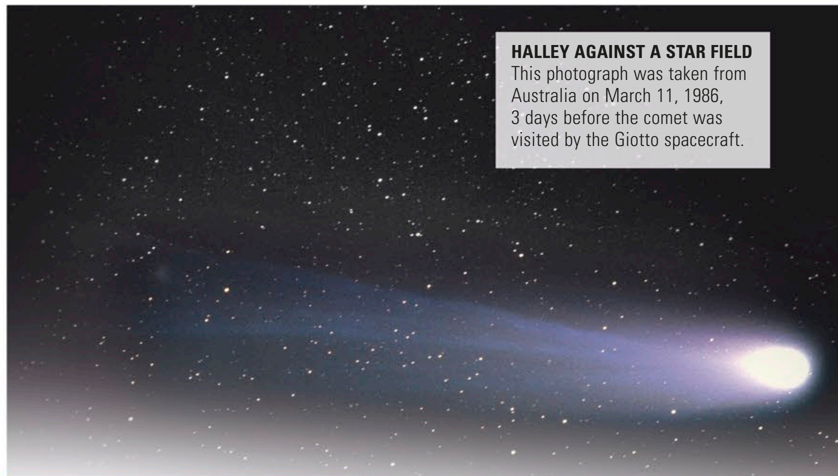
ORBITAL PERIOD 3.3 years

FIRST RECORDED January 17, 1786

Comet Encke was "discovered" in 1786 (by the French astronomer Pierre Méchain), in 1795 (by the German-born astronomer Caroline Herschel), and in 1805 and 1818–1819 (by the French astronomer Jean Louis Pons). These comets were found to be the same only after orbital calculations in 1819 by the German astronomer Johann Encke, who then predicted its return in 1822. Comet Encke is unusual in that, like Halley's Comet, it is not named after its discoverer. It has the shortest period of any known comet and has been seen returning to the Sun at intervals of 3.3 years since then. The orbit is also shrinking in size, as Encke comes back to perihelion about 2.5 hours sooner than it should. Some astronomers have suggested that this is due to the comet plowing through a resistive medium in the solar system. But other comets have returned later than predicted, and the time error has varied from one orbit to the next. Astronomers have realized that the changing orbits were caused by the "jet effect" of gas escaping from the comet's nucleus. The comet receives a push from the expanding gases and, depending on its direction of spin in relation to its orbit, it is either accelerated or decelerated.



COMA
Not all comets have tails. Some, such as Encke, often only have a dense spherical envelope of gas and dust around the nucleus called the coma. The density of the gas decreases as it flows away from the nucleus. Cometary comae have no boundaries; they just fade away.



HALLEY AGAINST A STAR FIELD
This photograph was taken from Australia on March 11, 1986, 3 days before the comet was visited by the Giotto spacecraft.

INTERMEDIATE-PERIOD COMET

Halley's Comet

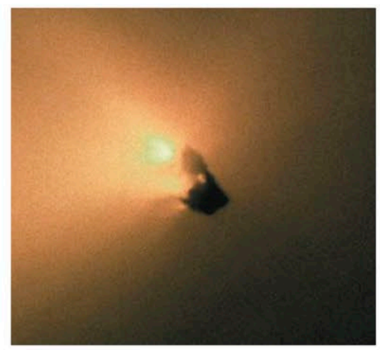
CLOSEST APPROACH TO THE SUN 55 million miles (88 million km)

ORBITAL PERIOD About 76 years

FIRST RECORDED 240 BCE

In 1696, Edmond Halley, England's second Astronomer Royal, reported to the Royal Society in London that comets that had been recorded in 1531, 1607, and 1682 had very similar orbits. He concluded that this was the same comet returning to the inner solar system about every 76 years, moving under the influence of the newly discovered solar gravitational force. What is more, Halley predicted that the comet would return in 1758. Halley's Comet was the first periodic comet to be discovered. This indicated that at least some comets were permanent members of the solar system.

Orbital analysis has revealed that Halley's Comet has been recorded 30 times, the first known sighting being in Chinese historical diaries of 240 BCE. The last appearance, in 1986, was 30 years after the start of the space age, and five spacecraft visited the comet. The most productive was ESA's Giotto mission. This flew to within 370 miles (600 km) of the nucleus and took the first-ever pictures. Giotto proved that cometary nuclei are large, potato-shaped dirty snowballs and that the majority of the snow is water ice. Halley was about 93 million miles (150 million km) from the Sun when Giotto encountered it. Only about 10 percent of the surface was actively emitting gas and dust at the time. On average, a comet loses a surface layer about 6.5 ft (2 m) deep every time it passes through the inner solar system. At this rate, Halley's Comet will survive for about another 200,000 years.



NUCLEUS
Giotto revealed that Halley's nucleus is 9.5 miles (15.3 km) long. The brightest parts of this image are jets of dust streaming toward the Sun.

MYTHS AND STORIES

CELESTIAL OMEN

Some superstitious people regard comets as portents of death and disaster. Before Edmond Halley's work, all comets were unexpected. They were often compared to flaming swords. England's King Harold II was worried by the appearance of Halley's Comet in 1066. But what was a bad omen for him was a good sign for the Norman Duke William, who conquered Harold at Hastings.

BAYEUX TAPESTRY
This crewel embroidery beautifully depicts the coma and tail of Halley's Comet (top left), as seen in 1066. It looks like a primitive rocket spewing out flames.